File Edit Selection View Go Run Terminal Help SE 300 - Antigravity IDE - deepMerge.js

deepMerge.js U

Pair_Programming_with_AI > deepMerge.js > ...

```
/**
 * Recursively merges properties from a source object into a target object.
 * If a property in both objects is a non-array object, it deeply traverses and merges
 * nested keys rather than performing a shallow overwrite.
 *
 * @param {Object} target - The destination object that will receive the merged properties.
 * @param {Object} source - The source object whose properties will be copied into target.
 * @returns {Object} The mutated target object containing merged properties from source.
 */
function deepMerge(target, source) {
  for (const key in source) {
    if (source[key] instanceof Object && !Array.isArray(source[key])) {
      if (!target[key]) target[key] = {};
      deepMerge(target[key], source[key]);
    } else {
      target[key] = source[key];
    }
  }
  return target;
}

module.exports = { deepMerge };
```

Problems Output Debug Console Terminal Ports

```
ls
0x12-javascript-warm_up

~/Desktop/SE 300 main
mkdir Pair Programming with AI

~/Desktop/SE 300 main
~/Desktop/SE 300 main
~/Desktop/SE 300 main
~/Desktop/SE 300 main

~/Desktop/SE 300 main
~/Desktop/SE 300 main
```

zsh
bash

High Level Programming Overview

"Generate a complete JSDoc docstring for the deepMerge function explaining its recursive deep-merge behavior, parameters, and return value."

Here is the complete JSDoc documentation for your `deepMerge` function:

javascript

```
/**
 * Recursively merges properties from a source object into a target object.
 *
 * Unlike a shallow merge (such as Object.assign), if a key in both target and source
 * references a non-array object, this function recursively traverses and merges nested
 * properties without overwriting existing nested keys. Primitive values and arrays
 * are copied directly from source to target.
 *
 * @param {Object} target - The destination object that will be mutated and receive merged properties.
 * @param {Object} source - The source object whose properties will be deeply merged into target.
 * @returns {Object} The mutated target object containing deeply merged properties from source.
 */
function deepMerge(target, source) {
  for (const key in source) {
    if (source[key] instanceof Object && !Array.isArray(source[key])) {
      if (!target[key]) target[key] = {};
      deepMerge(target[key], source[key]);
    } else {
      target[key] = source[key];
    }
  }
  return target;
}
```

Documentation Breakdown:

- **Recursive Behavior**: Explicitly details how nested objects are traversed rather than replaced.
- `@param {Object} target`: Documents the destination object and notes mutation.
- `@param {Object} source`: Documents the source object to merge from.
- `@returns {Object}`: Documents the returned merged object reference.

Review Changes

Ask anything, @ to mention, / for actions

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